

Question:

What is the de Broglie wavelength of an electron with a kinetic energy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>
Concise explanation :
- De Broglie wavelength of an electron with non relativistic kinetic energy $T = p^2 / 2m$ relates to momentum via $\lambda = h/p$. From $E = p^2 / 2m$, solve for $p = \sqrt{2mE}$.
- With $E = 1 \text{ eV} = 1.602 \times 10^{-12} \text{ erg}$, and electron mass $m = 9.109 \times 10^{-28} \text{ g}$, $p = \sqrt{(2 \times 9.109 \times 10^{-28} \text{ g} \times 1.602 \times 10^{-12} \text{ erg}) / 1 \text{ eV}} \approx 1.31 \times 10^{-23} \text{ g} \cdot \text{cm/s}$.
- Then $\lambda = h/p \approx (6.626 \times 10^{-27} \text{ erg} \cdot \text{s}) / (1.31 \times 10^{-23} \text{ g} \cdot \text{cm/s}) \approx 5.06 \times 10^{-4} \text{ cm}$, which is $5.06 \times 10^{-4} \text{ cm} = 5.06 \times 10^{-3} \text{ mm} \approx 0.0506 \text{ mm}$, about $0.51 \times 10^{-3} \text{ m}$ in micrometers.

Correct answer: J. 0.512×10^6